



**BAULKHAM HILLS
HIGH
SCHOOL**

2022

**YEAR 12
H3: ASSESSMENT
TASK 3**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In all questions, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

**Total
marks:
40**

- There are 4 Questions in total
- All questions should be attempted

40 marks

Attempt Questions 1 – 4

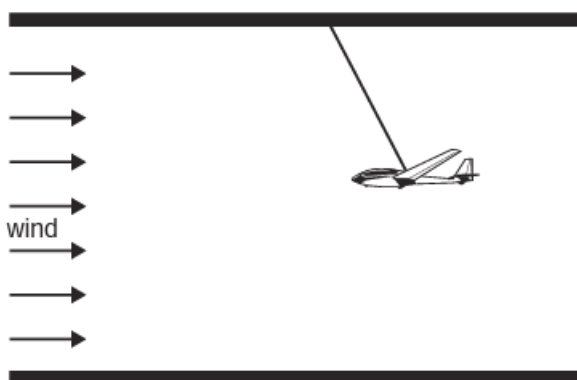
Answer each question on the appropriate answer sheet. Each answer sheet must show your NES A#. Extra paper is available.

For all questions your responses should include relevant mathematical reasoning and/or calculations.

Question 1 (9 marks) Use the pages labelled Question 1 in the answer booklet

- (a) A model glider with a mass of 4 kg is suspended from the roof of a wind tunnel by a thin wire. A light wind exerts a horizontal force of magnitude $\frac{g}{2}$ Newtons on the model glider.

- (i) Let T Newtons be the magnitude of the tension in the suspending wire. 1
Clearly label all forces acting on the model glider on the copy of the below diagram on your answer sheet.



- (ii) Calculate the value of T , giving your answer in the form $\frac{g\sqrt{a}}{b}$ where a and b 2
are positive integers.
- (b) The acceleration of a particle P moving in a straight line is given by

$$\ddot{x} = 3x(x - 4)$$

where x metres is the displacement from the origin O and t is the time in seconds.
Initially the particle is at O and its velocity is $4\sqrt{2}$ m/s.

- (i) Show that $v^2 = 2(x^3 - 6x^2 + 16)$, where v is the velocity of P . 2
- (ii) Calculate the velocity and acceleration of P at $x = 2$. 2
- (iii) In which direction does P move from $x = 2$? Give a reason for your answer. 1
- (iv) Briefly describe the motion of P after it moves from $x = 2$. 1

Question 2 (11 marks) Use the pages labelled Question 2 in the answer booklet

(a) The position vectors of points P and Q are $\underline{p} = 2\underline{i} - 3\underline{j} + \underline{k}$ and $\underline{q} = 2\underline{i} + 2\underline{j} - 4\underline{k}$ respectively.

(i) Find the magnitude of $\underline{p}$. 1

(ii) Determine $\angle POQ$, to the nearest degree, where O is the origin. 2

(iii) Find the vector equation of the line joining P and Q . 2

(iv) Which plane does the vector $\overrightarrow{PQ}$ lie on? 1

(v) Find a vector equation for a line that would intersect $\overrightarrow{PQ}$ and is also perpendicular to $\overrightarrow{PQ}$. 2

(vi) Find the position vector of any point that lies on your line in part (v), but does not lie on $\overrightarrow{PQ}$. 1

(b) What is the Cartesian equation of the line $\underline{r} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -3 \end{pmatrix}$? 2

Question 3 (10 marks) Use the pages labelled Question 3 in the answer booklet

- (a) A body falling from rest experiences resistance directly proportional to its velocity squared. (i.e. resistive force = mkv^2 where k is a constant)

(i) Write the equation of motion for this body. 1

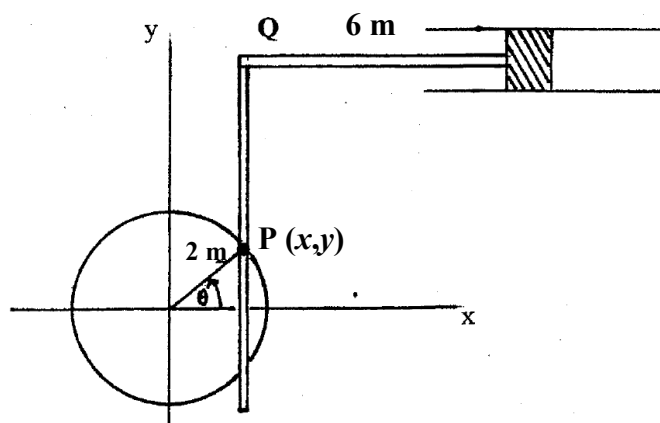
(ii) Show that the distance fallen when the velocity is V is given by 2

$$x = \frac{1}{2k} \ln \left| \frac{g}{g - kV^2} \right|$$

(iii) Calculate the terminal velocity. 1

- (b) A piston moves horizontally back and forth on the end of a 6 metre shaft.

The other end is attached at Q to a vertical slotted arm fitted to a peg P on the rim of a wheel of radius 2 metres.



Suppose the wheel begins with point P at $\theta = \frac{\pi}{4}$, when $t = 0$ and rotates anticlockwise at 5 radians per second.

(i) Show that $x = 2\cos\left(5t + \frac{\pi}{4}\right)$. 2

(ii) Show that the motion of the piston is simple harmonic. 1

(iii) State the amplitude and period of the motion 2

(iv) Find the initial velocity of the piston. 1

Question 4 (10 marks) Use the pages labelled Question 4 in the answer booklet

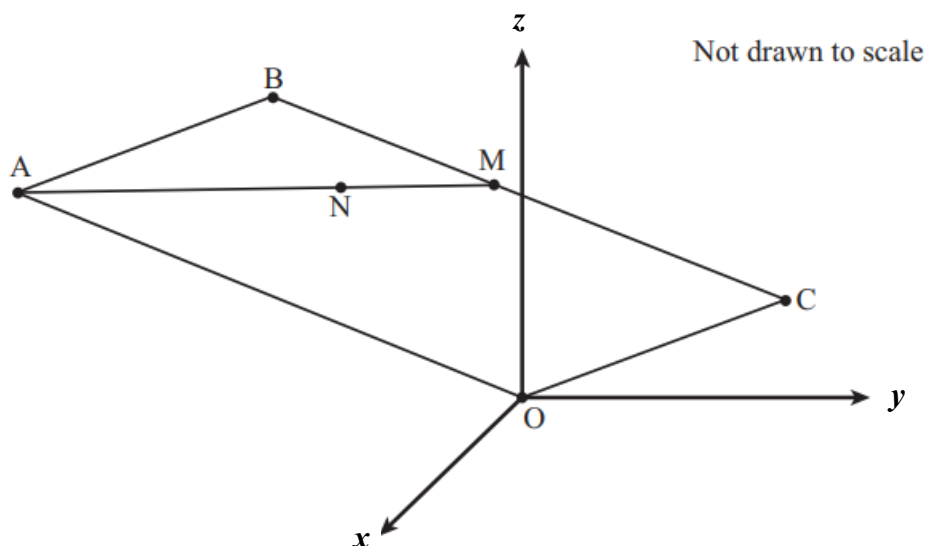
- (a) Consider two lines in three dimensions given by

3

$$\vec{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \quad \text{and} \quad \vec{r} = \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}$$

By equating components, find the point of intersection of the two lines.

- (b) The points $A(-6,2,-2)$ and $C(3,1,2)$ are represented in three-dimensional space in the diagram

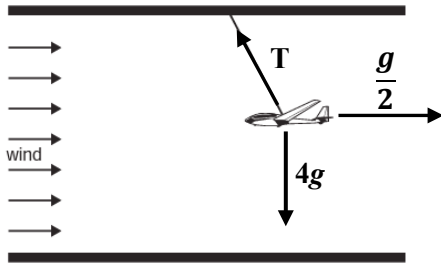
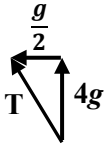


$OABC$ forms a parallelogram in three-dimensional space. Using vector techniques;

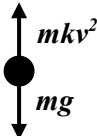
- (i) Determine the coordinates of B . 2
- (ii) M is the midpoint of BC . Determine the vector that represents $\vec{MO}$. 1
- (iii) N divides AM in the ratio 2:1. Determine the vector that represents $\vec{ON}$. 2
- (iv) Show that O , B and N lie on a straight line. 2

End of paper

BAULKHAM HILLS HIGH SCHOOL
YEAR 12 EXTENSION 2 H3: ASSESSMENT TASK 3 2022 SOLUTIONS

Solution	Marks	Comments
QUESTION 1		
1 (a) (i) 	1	1 mark Correct answer with all three forces shown with direction indicated
1 (a) (ii)  As the model is in equilibrium, both the horizontal and vertical forces must balance. $T^2 = \left(\frac{g}{2}\right)^2 + (4g)^2$ $= \frac{g^2}{4} + 16g^2$ $= \frac{65g^2}{4}$ $T = \frac{g\sqrt{65}}{2} \text{ Newtons}$	2	2 marks - Correct solution 1 mark - Attempts to resolve forces in a correct approach
1 (b) (i) $v \frac{dv}{dx} = 3x(x - 4)$ $\int_{4\sqrt{2}}^v v dv = \int_0^x (3x^2 - 12x) dx$ $\left[\frac{v^2}{2} \right]_{4\sqrt{2}}^v = 2 \left[x^3 - 6x^2 \right]_0^x$ $v^2 - 32 = 2(x^3 - 6x^2)$ $v^2 = 2(x^3 - 6x^2 + 32)$	2	2 marks - Correct solution 1 mark - Uses $\ddot{x} = v \frac{dv}{dx}$ and separates the variables
1 (b) (ii) When $x = 2$; $v^2 = 2(2^3 - 6(2)^2 + 16)$ $= 0$ $v = 0$ $\ddot{x} = 3(2)(2 - 4)$ $= -12$	2	2 marks - Correct solution 1 mark - Finds either velocity or acceleration
1 (b) (iii) As $v = 0$ and $\ddot{x} < 0$, the particle must move towards the origin.	1	1 mark Correct answer with explanation
1 (b) (iv) After the particle passes through the origin it will slow down, as $v < 0$ and $\ddot{x} > 0$, until it stops and returns to the origin	1	1 mark A description of the motion that explains why the particle is slowing down.

Solution	Marks	Comments
QUESTION 2		
2 (a) (i) $ \tilde{p} = \sqrt{2^2 + 3^2 + 1^2}$ $= \sqrt{14}$	1	1 mark • Correct answer
2 (a) (ii) $\cos \angle POQ = \frac{\tilde{p} \cdot \tilde{q}}{ \tilde{p} \tilde{q} }$ $= \frac{4 - 6 - 4}{\sqrt{14} \sqrt{24}}$ $= -\frac{6}{\sqrt{336}}$ $\angle POQ = 109^\circ$ (to nearest degree)	2	2 marks • Correct solution 1 mark • Finds $\tilde{p} \cdot \tilde{q}$
2 (a) (iii) $\tilde{r} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2-2 \\ 2+3 \\ -4-1 \end{pmatrix}$ $= \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 5 \\ -5 \end{pmatrix}$ $= \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$	2	2 marks • Correct solution 1 mark • Finds a correct direction vector
2 (a) (iv) $\overrightarrow{PQ}$ lies on the plane $x = 2$	1	1 mark • Correct answer
2 (a) (v) let the direction vector of the line be $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$. $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 0$ $y - z = 0$ $y = z$ $\therefore$ a possible equation would be $\tilde{r} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$	2	2 marks • Correct solution 1 mark • Uses $\tilde{p} \cdot \tilde{q} = 0$ to establish perpendicular lines <i>Note: point must lie on $\overrightarrow{PQ}$</i>
2 (a) (vi) let $\mu = 1$ $\tilde{r} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 2 \\ -2 \\ 2 \end{pmatrix}$	1	1 mark • Correct answer
2 (b) required slope $= -\frac{3}{6}$ $= -\frac{1}{2}$	2	2 marks • Correct solution 1 mark • Finds the slope of the line • Attempts to eliminate the parameter

Solution		Marks	Comments
QUESTION 3			
3 (a) (i)	 $m\ddot{x} = mg - mkv^2$ $\ddot{x} = g - kv^2$	1	1 mark • Correct answer
3 (a) (ii)	$v \frac{dv}{dx} = g - kv^2$ $\int_0^v \frac{v}{g - kv^2} dv = \int_0^x dx$ $x = -\frac{1}{2k} [\ln g - kv^2]_0^v$ $= -\frac{1}{2k} \ln \left \frac{g - kv^2}{g} \right $ $= \frac{1}{2k} \ln \left \frac{g}{g - kv^2} \right $	2	2 marks • Correct solution 1 mark • Separates the differential equation into two valid integrals
3 (a) (iii)	terminal velocity occurs when $\ddot{x} = 0$ i.e. $g - kv^2 = 0$ $v^2 = \frac{g}{k}$ $v = \sqrt{\frac{g}{k}}$	1	1 mark • Correct answer
3 (b) (i)	$\frac{d\theta}{dt} = 5$ $\int_{\frac{\pi}{4}}^{\theta} d\theta = 5 \int_0^t dt$ $\theta - \frac{\pi}{4} = 5t$ $\theta = 5t + \frac{\pi}{4}$ $\frac{x}{2} = \cos\theta$ $x = 2\cos\theta$ $x = 2\cos\left(5t + \frac{\pi}{4}\right)$	2	2 marks • Correct solution 1 mark • Establishes $\theta = 5t + \frac{\pi}{4}$
3 (b) (ii)	$x = 2\cos\left(5t + \frac{\pi}{4}\right)$ $\dot{x} = -10\sin\left(5t + \frac{\pi}{4}\right)$ $\ddot{x} = -50\cos\left(5t + \frac{\pi}{4}\right)$ $\ddot{x} = -25x$ ∴ motion is SHM as it takes the form $\ddot{x} = -n^2x$	1	1 mark • Correct solution
3 (b) (iii)	amplitude = 2 metres period = $\frac{2\pi}{n}$ $= \frac{2\pi}{5} \text{ seconds}$	2	2 marks • Correct solution 1 mark • Correctly finds either amplitude or acceleration
3 (b) (iv)	when $t = 0$; $\dot{x} = -10\sin\frac{\pi}{4}$ $= -10\left(\frac{1}{\sqrt{2}}\right)$ $= -5\sqrt{2} \text{ m/s}$	1	1 mark • Correct answer

Solution	Marks	Comments
QUESTION 4		
4 (a) equating the z components: $3 + 4\lambda = \mu$ equating the y components: $2 + 3\lambda = 1 + 2\mu$ $= 7 + 8\lambda$ $5\lambda = -5$ $\lambda = -1$ $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$ $= \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}$ $\therefore$ the points meet at $(-1, -1, -1)$	3	3 marks - Correct solution <i>Note: answer must be a point NOT a vector</i> 2 marks - Finds either λ or μ 1 mark - Establishes a relationship between λ and μ
4 (b) (i) $\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}$ $= \overrightarrow{OA} + \overrightarrow{OC}$ $= \begin{pmatrix} -6 \\ 2 \\ -2 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} -3 \\ 3 \\ 0 \end{pmatrix}$ $\therefore$ B is the points meet at $(-3, 3, 0)$	2	2 marks - Correct solution 1 mark - Uses a correct technique
4 (b) (ii) $\overrightarrow{MO} = -\frac{1}{2}(\overrightarrow{OB} + \overrightarrow{OC})$ $= -\frac{1}{2}\left[\begin{pmatrix} -3 \\ 3 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}\right]$ $= \begin{pmatrix} 0 \\ -2 \\ -1 \end{pmatrix}$	1	1 mark Correct answer
4 (b) (iii) $\overrightarrow{ON} = \overrightarrow{OA} + \frac{2}{3}\overrightarrow{AM}$ OR $\overrightarrow{ON} = \frac{1}{3}\begin{pmatrix} -6 \\ 2 \\ -2 \end{pmatrix} + \frac{2}{3}\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} -6 \\ 2 \\ -2 \end{pmatrix} + \frac{2}{3}\left[\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} - \begin{pmatrix} -6 \\ 2 \\ -2 \end{pmatrix}\right]$ $= \begin{pmatrix} -6 \\ 2 \\ -2 \end{pmatrix} + \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix}$ $= \begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix}$	2	2 marks - Correct solution 1 mark - Finds an expression for $\overrightarrow{ON}$ involving other vectors
4 (b) (iv) $\overrightarrow{ON} = \begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix}$ $\overrightarrow{OB} = \begin{pmatrix} -3 \\ 3 \\ 0 \end{pmatrix}$ $= \frac{2}{3}\overrightarrow{OB}$ $\therefore \overrightarrow{ON} \parallel \overrightarrow{OB}$ as they are scalar multiples of each other However as the two vectors share the common point O, O, B and N must be collinear.	2	2 marks - Correct solution 1 mark - Shows that two relevant vectors are parallel